UK Online Retail Sales Analysis

SQL Data Analysis & Power BI Business Intelligence Dashboard

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Tools Used: PostgreSQL, SQL, Power BI

Dataset: UK Online Retail Dataset (2010–2011)

1. Introduction

This project analyses transactional data from a UK-based online retail company between December 2010 and December 2011. The goal of the analysis is to understand business performance, identify customer purchasing patterns, evaluate product performance, and uncover geographic sales trends.

The project combines SQL-based data analysis with Power BI dashboards to transform raw transactional data into meaningful business insights.

2. Data Preparation

The dataset was first imported into PostgreSQL using a staging table where all fields were initially stored as text to prevent ingestion errors. A cleaned analytical view was then created to standardize data types and calculate revenue.

Cleaning steps included:
- Converting quantity and price to numeric types
- Converting invoice dates to timestamp
- Creating a calculated revenue column (quantity × unit_price)
- Removing cancelled orders and invalid transactions

3. Core Business KPIs

Key performance metrics were calculated using SQL aggregation queries after filtering out transactions with missing CustomerID values.

Metric	Value
Total Revenue	€8,911,407.90
Total Orders	18,532
Active Customers	4,338
Average Order Value (AOV)	€480.87

4. Sales Trend Analysis

Monthly sales analysis shows clear seasonal patterns. Revenue remained relatively stable during the first half of the year before accelerating significantly in Q4. November represents the peak month, likely driven by pre-Christmas wholesale purchasing.

5. Customer Behaviour Analysis

Customer segmentation revealed strong repeat purchasing behaviour. Approximately 66% of customers are repeat buyers, indicating high retention. The distribution of order frequency shows that most customers place between two and six orders, while a smaller group of high-frequency customers place many more.

6. Product Performance

Product analysis revealed that a relatively small number of items generate the majority of revenue. Some products achieve high revenue through high pricing while others generate revenue through high sales volume. This indicates opportunities for pricing optimization and product promotion.

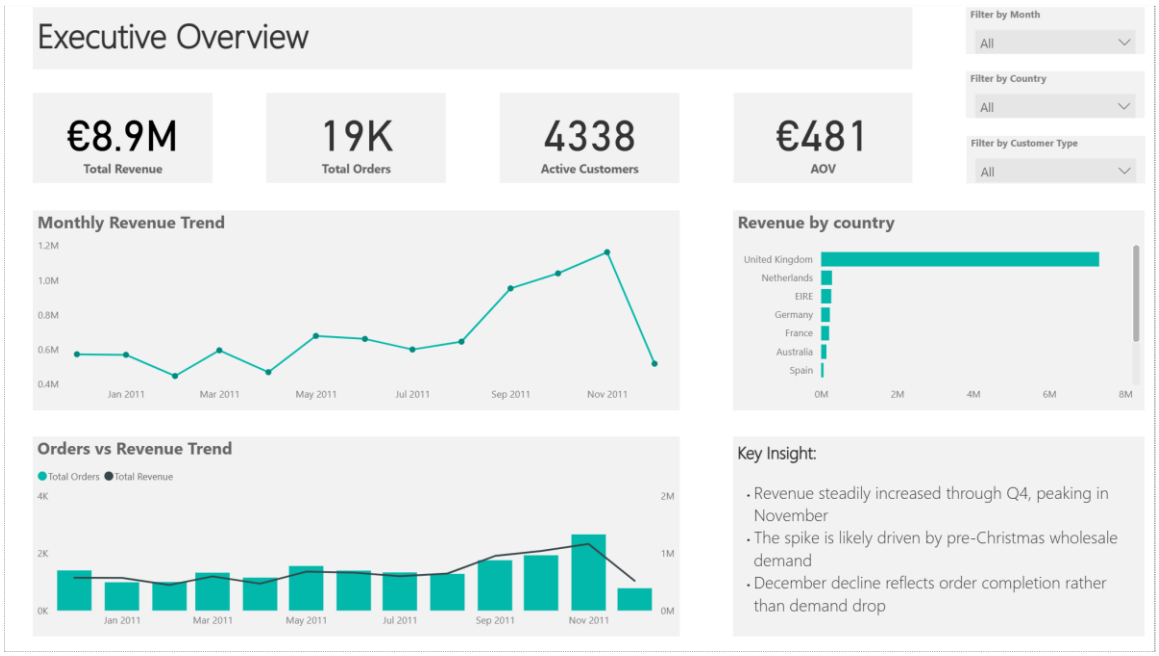
7. Geographic Sales Analysis

The United Kingdom dominates both revenue and order volume. However, several international markets demonstrate higher average order values, including the Netherlands and Singapore. These regions may represent opportunities for international growth.

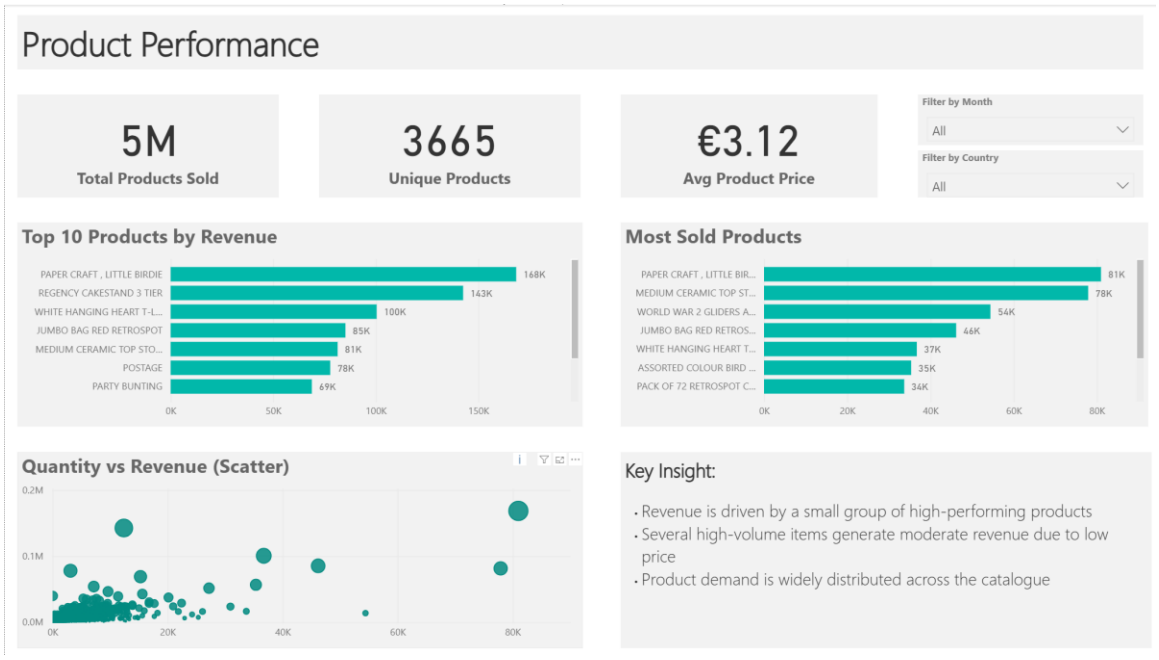
8. Power BI Dashboard Overview

The analysis results were visualized in Power BI using four dashboard pages:

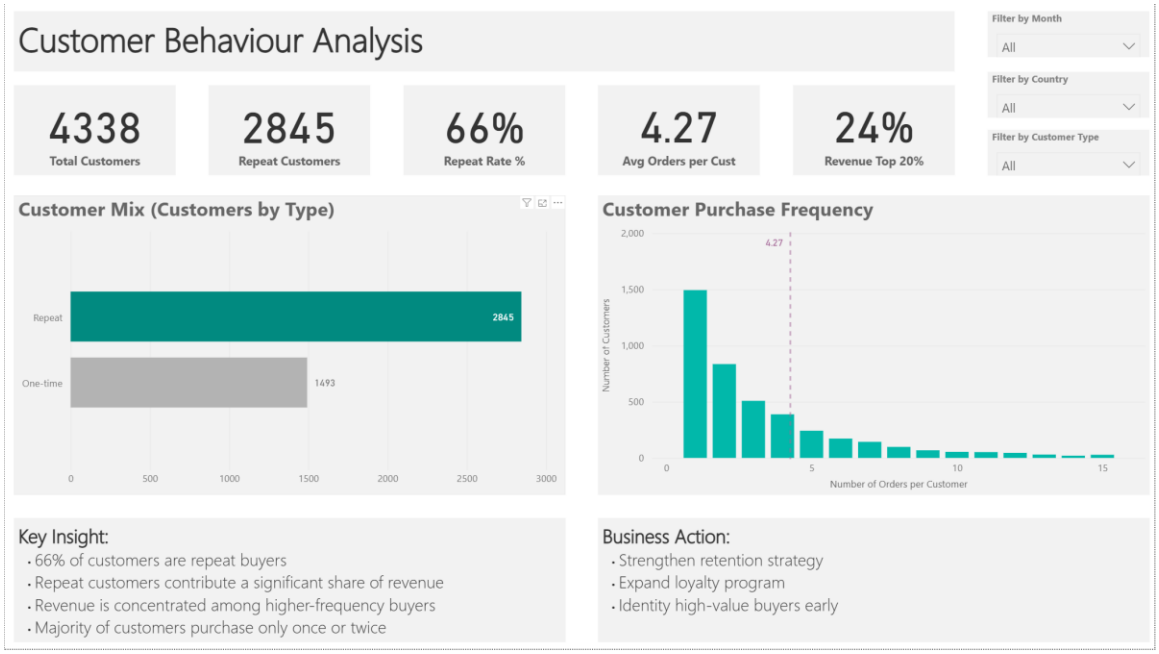
- Executive Overview: high-level KPIs and revenue trends



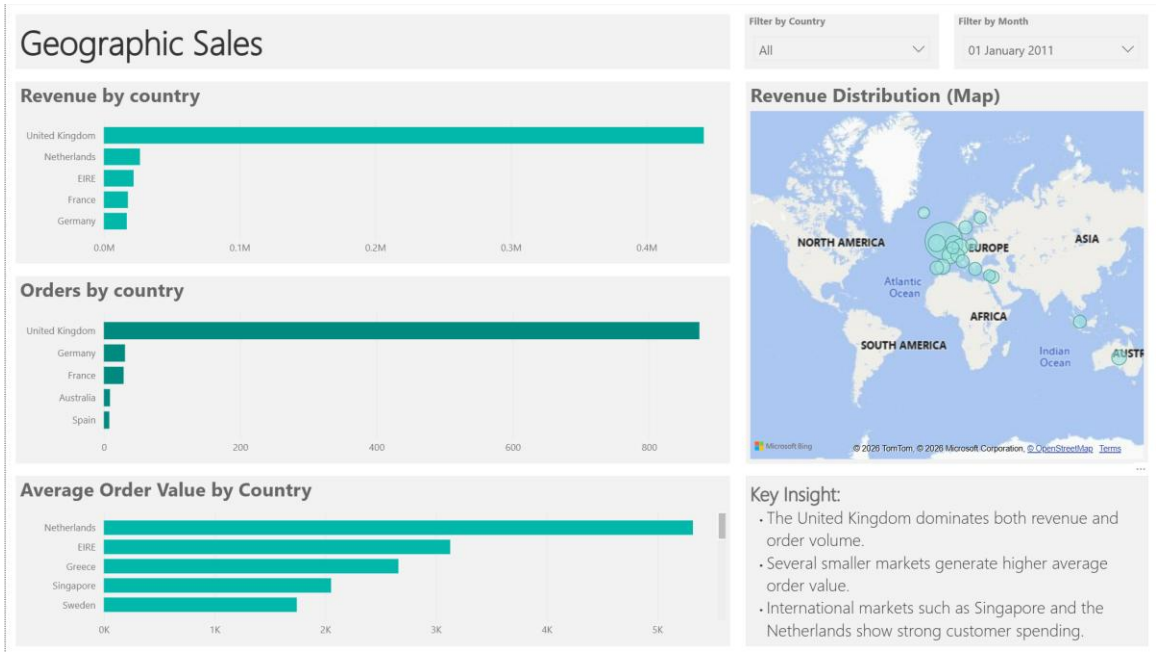
- Product Performance: top revenue products and sales distribution



- Customer Behaviour: repeat rate and purchase frequency



- Geographic Sales: country-level revenue and map visualisations



9. Key Business Insights

- Revenue growth is strongest during Q4
- A large share of revenue comes from repeat customers
- A small number of products drive a large share of total revenue
- International markets show strong spending potential

10. Business Recommendations

- Strengthen customer retention programs
- Focus marketing on high-performing products
- Expand marketing efforts in high-value international markets
- Prepare inventory and operations for seasonal demand spikes

11. Conclusion

This project demonstrates a full analytics workflow from SQL data preparation and business analysis to interactive Power BI dashboards. The insights produced can support data-driven decision making and strategic planning.

12. Appendix: Example SQL KPI Query

```
--1. Total Revenue

SELECT ROUND(

SUM(line_revenue)) AS total_revenue

FROM v_retail_clean

WHERE customer_id IS NOT NULL;
```

	total_revenue numeric
1	8911408

```
--2. Total Orders

SELECT

COUNT(DISTINCT invoice_no) AS total_orders

FROM v_retail_clean

WHERE customer_id IS NOT NULL;
```

	total_orders bigint
1	18532

```
--3. Active Customers

SELECT

COUNT(DISTINCT customer_id) AS active_customers

FROM v_retail_clean

WHERE customer_id IS NOT NULL;
```

	active_customers bigint
1	4338

```
--4. Average Order Value (AOV)

SELECT ROUND(
```

```
SUM(line_revenue) / NULLIF(COUNT(DISTINCT invoice_no),0)) AS avg_order_value

FROM v_retail_clean

WHERE customer_id IS NOT NULL;
```

	avg_order_value numeric
1	481